MIL-C-60275A(MU)
5 February 1971
SUPERSEDING
MIL-C-60275(MU)
27 September 1966

MILITARY SPECIFICATION

CARTRIDGE, 20MM, TARGET PRACTICE, M204

1. SCOPE

1.1 This specification covers one type of percussion primed cartridge for use in Gun, 20mm, Automatic, M3.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of the specification to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-644 - Visual Inspection Standards and Inspection Procedures for Inspection of Packaging, Packing and Marking of Small Arms Ammunition
MIL-STD-651 - Visual Inspection Standards for 20mm Ammunition and Components
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

U.S. ARMY MUNITIONS COMMAND

D7258928 - Marking on Shipping and Storage Container, M548, for 20mm Cartridges
F7258967 - Method of Packing 150 20mm Cartridges, in M10 Links, Three 50-round Belts, in M548 Container; Packed for Single Loop Right Hand Feed and Double Loop Left Hand Feed

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2.1 DRAWINGS (cont)

D7259106	- Cartridge, 20mm, Target Practice, M204
IEL 7259106	- Index of Inspection Equipment Lists for: Cartridge, 20mm, TP, M204
F7259108	- Method of Bulk Packing 200 Cartridges, 20mm, Target Practice, with M21A1 Percussion Primed Case, in M548 Container
F7259123	- Method of Bulk Packing 150 Cartridges, 20mm, Target Practice, with M21A1 Type Cases, in M548 Container
F7259188	- Method of Packing 150 20mm Cartridges, in M10 Link, Three 50-round Belts, in M548 Container; Packed for Single-Loop Left Hand Feed and Double-Loop Right Hand Feed
F7259189	- Method of Packing 150 20mm Cartridges, in M10 Link, One Belt, in M548 Container; Packed for Single-Loop Left Hand Feed and Double-Loop Right Hand Feed
D7259237	- Marking on Shipping and Storage Container, M548 for Cartridges, 20mm, Linked (Functional)
F11075755	- Method of Packing 150 20mm Cartridges, in M10 Links, One Belt, in M548 Container; Packed for Single-Loop Right Hand Feed and Double-Loop Left Hand Feed

PUBLICATIONS

U.S. ARMY MATERIEL COMMAND

AMCR 715-505 - Ammunition Ballistics Acceptance Test Methods, Volume 8, Test Procedures for 20mm Cartridges

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions shall be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.-The cartridge shall comply with Drawing (Dwg.) D7259106, referenced specifications, and other requirements specified herein.

3.2 Manufacturing process.-All parts and assemblies shall be manufactured by a process approved by the contracting officer, and no deviation from that process shall be made without his prior approval. (see 6.3)

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3.3 Projectile torque.-The projectile shall withstand a minimum torque, without rotary movement, as specified by Dwg. D7259106.

3.4 Projectile extraction.-The projectile shall separate from the cartridge case within tension limits specified on Dwg. D7259106.

3.5 Velocity.-The average velocity of projectiles of the sample cartridges conditioned at 68 degrees Fahrenheit (^{o}F) to 72 ^{o}F shall be 2680 ± 50 feet per second (fps) at 78 feet from the weapon muzzle. The sample standard deviation of the velocities shall not exceed 40 fps.

3.6 Pressure.-The average chamber pressure ($\overline{X}$) of the sample cartridges conditioned at 68 ^{o}F to 72 ^{o}F, plus 0.84 times the average range ($\overline{R}$) of the chamber pressure, shall not exceed 51,000 pounds per square inch (psi).

3.7 Waterproofness.-The average velocity (wet) of projectiles of the sample cartridges conditioned at 68 ^{o}F to 72 ^{o}F shall not vary from the average velocity (dry) by more than 100 fps.

3.8 Accuracy.-The average mean radius of all targets resulting from the sample cartridge fired at ambient temperature shall not exceed 15 inches at a range of 600 yards.

3.9 Action time.-The action time of the sample cartridges shall not exceed 4.0 milliseconds at ambient temperature.

3.10 Function and Casualty.-The sample cartridges, at ambient temperature, shall function without casualty.

3.11 Workmanship.-The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.11.1 Process defects.-The cartridge shall be free of defects such as folds, wrinkles, buckles, bulges, deep scratches, cuts, dents, perforations and other forms of substandard workmanship.

3.11.2 Cleanliness.-The cartridge and its components shall be free of all foreign matter. (see 6.5)

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirements.

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4.1.1 Quality assurance terms and definitions.-Reference shall be made to MIL-STD-109 for definition of quality assurance terms.

4.2 First article sample.

4.2.1 Initial production sample.-At the beginning of regular production a sample shall be submitted in accordance with contract requirements and shall consist of 380 cartridges. (see 6.2) The sample shall be manufactured using the same materials, equipment, processes and procedures as will be used in regular production. Identification shall be in accordance with MIL-STD-1168.

4.2.1.1 Examination and test.-After inspection and provisional acceptance at source the sample will be inspected for all requirements of the drawings and specifications at a Government laboratory or other such facility specified by the contract. (see 6.2)

4.2.1.2 Initial production sample failure.-Failure of the sample to comply with requirements of the drawings and specifications will result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Submission of product.-The product shall be submitted in accordance with MIL-STD-105 or MIL-STD-1235, as applicable.

4.3.1.1 Lot.-A lot shall consist of:

a. Cartridges loaded by one manufacturer, in one unchanged process, in accordance with the same drawings and drawing revisions, and same specification and specification revision.

b. Like parts and assemblies (cases, primers, projectiles) having one manufacturer's symbol and one interfix number.

c. Propellant from one lot.

4.3.1.2 Lot identification.-Each cartridge and each packed ammunition lot shall be identified in accordance with applicable drawings and MIL-STD-1168. Each packed lot shall be further identified by a federal stock number assigned by the procuring activity.

4.3.2 Examination.-One hundred percent examination shall be performed by the contractor for all critical defects. Examination for major and minor defects shall be performed as specified herein. Any cartridge found to be defective shall be rejected.

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4.3.2.1 Sampling and acceptance criteria.-Except as otherwise indicated, sampling plans and procedures shall be in accordance with MIL-STD-105 or MIL-STD-1235. Defect characteristics may be grouped into a class and considered collectively, or defect characteristics may be considered individually to determine acceptability. When acceptability is determined on a class basis, in addition to the acceptable quality level (AQL) assigned to the major class, an AQL shall be assigned to individual characteristics within the major class, limiting the number of defectives containing a particular major defect to less than the number of defectives permitted for the class. However, where sampling procedures of MIL-STD-1235 are used, acceptability shall not be determined on a class basis for major defects. (see 6.4)

4.3.2.2 AQL's.-The following AQL's are assigned:

a. Individual defect characteristic basis.

Major defects ---------- 0.25%
Minor defects ---------- 0.40%

b. Defect characteristic, class basis.

Major defects ---------- 1.50%
Minor defects ---------- 2.50%

c. Individual defect characteristic associated with the class basis.

Major defects ---------- 0.40%

4.3.3 Classification of defects 1/.-The classification of defects shall be as follows:

4.3.3.1 Cartridge assembly (see Dwg. D7259106).

Category and Defects	Method of Inspection
Critical:	
1. Crack or split in projectile body	Visual
2. Split or perforated case	Visual
3. Primer above flush of case head face	Gage
Major:	
101. Cocked, loose, missing or inverted primer	Visual/Manual
102. Depth of primer seating	Gage
103. Overall length, max.	Gage
104. Improper, incomplete, or missing crimp (case-projectile)	Visual
105. Profile and alignment, max.	Gage 2/

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4.3.3.1 Cartridge assembly (see Dwg. D7259106)(cont).

Category and Defects	Method of Inspection
Major: (cont)	
106. Corrosion	Visual
107. Head configuration	Visual
Minor:	
201. Marking incorrect, incomplete, illegible or missing	Visual
202. Projectile protective coating damaged or incomplete	Visual
203. Waterproofing missing around primer	Visual
204. Scratch, dent, buckle, bulge, wrinkle or fold in case	Visual
205. Rotating band damaged	Visual
206. Primer crimp missing	Visual
207. Foreign matter, except corrosion	Visual
208. Incorrect type cartridge	Visual

4.3.3.2 Cartridge assembly after application of wax coating (see Dwg. D7259106).

Category and Defects	Method of Inspection
Critical: None defined.	
Major:	
101. Wax coating missing or incomplete	Visual
Minor:	
201. Wax coating flaky or otherwise unsatisfactory	Visual
202. Wax coating on projectile	Visual

1/ MIL-STD-651 shall apply in defining and evaluating visual defects.

2/ A dead weight load of 60 pounds maximum may be used to insert the cartridge in the gage.

4.3.4 Tests.-The tests listed in Table I shall be performed on each cartridge lot in accordance with the test methods prescribed in 4.4. Sample size and acceptance criteria for each test shall be as specified. Only cartridges having met the visual and dimensional requirements and having been selected in such a manner that the sample is representative of the entire lot shall be used in the tests. The selected cartridges shall be thoroughly mixed before being divided into samples for the various tests.

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Table I

Test 1/		Sample size Production Lot	Rqmt Para
Projectile torque	2/	20	3.3
Projectile extraction	3/	20	3.4
Propellant contamination	4/	20	3.11.2
Velocity	5/	20	3.5
Pressure	6/	20	3.6
Waterproofness	7/	20	3.7
Accuracy	8/	50	3.8
Action time	9/	50	3.9
Function and casualty	10/	200	3.10

1/ The lot shall be rejected if in any firing test one or more of the following malfunctions or casualties of Table II occur:

a. Projectile remaining in bore.
b. Metal parts separation, except rotating band separation (see Table II, Note 4/).

Except as otherwise specified and in the function and casualty test where the occurrence of a firing defect listed in Table II prevents the obtaining of a reliable result for the characteristic being tested, an additional shot shall be fired.

2/ Projectile torque.-Failure of two or more of the sample cartridges to comply with the specified torque requirement shall cause rejection of the lot. If one cartridge of the sample fails to comply, a second sample of 20 cartridges shall be tested. The lot shall be rejected if two or more cartridges of the combined samples fail to comply with the applicable requirement.

3/ Projectile extraction.

a. Failure of two or more sample cartridges to comply with the minimum requirement shall cause rejection of the lot. If one cartridge of the sample fails to comply with the minimum requirement, a second sample of 20 cartridges shall be tested. The lot shall be rejected if two or more cartridges of the accumulated samples fail to comply with the minimum requirement.

b. Failure of four or more cartridges to comply with the maximum requirement shall cause rejection of the lot. If more than one but less than four cartridges fail to comply with the maximum requirement, a second sample of 20 cartridges shall be tested. The lot shall be rejected if five or more cartridges of the accumulated samples fail to comply with the maximum requirement.

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4/ Propellant contamination.-The presence of foreign matter in the propellant of any of the test cartridges shall cause rejection of the lot. (see 3.11.2 and 6.5)

5/ Velocity.-If the sample fails to comply with either or both requirements, a retest sample of 40 cartridges shall be tested for the failing parameter(s) (average, sample standard deviation). The lot shall be rejected if the retest sample fails to comply with the applicable requirements. (see 6.6.1)

6/ Pressure.-If the average pressure ($\bar{X}$) plus 0.84 times the average range ($\bar{R}$) of the sample cartridges exceeds 51,000 psi, a retest sample of 40 cartridges shall be tested. The lot shall be rejected if the average pressure ($\bar{X}$) plus 0.91 times the average range ($\bar{R}$) of the retest sample fails to comply with the applicable requirement. (see 6.6.2)

7/ Waterproofness.-If the average velocity (wet) of the sample fails to comply with the applicable requirement, a retest sample of 40 cartridges shall be tested. The lot shall be rejected if the average velocity (wet) differs by more than 100 fps from the average velocity (dry) of 40 additional sample cartridges fired to establish a new base for comparison. The lot shall not be penalized for failure of the last named sample cartridges to comply with the requirement of 3.5.

8/ Accuracy.-If the average mean radius of the sample cartridge projectiles exceeds the applicable requirement, a retest sample of 100 cartridges shall be tested. The lot shall be rejected if the average mean radius of 10 retest accuracy targets exceeds the applicable requirements.

9/ Action time.-If any sample cartridge fails to meet the requirement for action time, the lot shall be rejected.

10/ Function and casualty.-The lot shall be rejected if any firing defect of Table II occurs in number(s) equal to or greater than the applicable "REJ" number. Except as otherwise provided, if firing defects occur in excess of the applicable "ACC" number, but less than the applicable "REJ" number, a second sample of 400 cartridges shall be tested. The lot shall be rejected if in the accumulated first and second samples firing defects occur in numbers equal to or greater than the applicable "REJ" number.

4.3.4.2 Firing defects.-Firing defects and acceptance criteria shall be as specified in Table II. For defect definitions, see AMCR 715-505, Volume 8.

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Table II

Firing Defects	ACC	REJ
Cartridge:		
Misfire	0	2
Failure to chamber	0	1
Failure to extract	0	1
Projectile remaining in bore	0	1
Primer:		
Primer leak	9	25
Primer perforation	1	6
Loose primer	0	1
Blown primer	0	1
Case:		
Longitudinal split 1/:		
I or S	10	31
J	4	11
K	0	2
L or M	0	1
Circumferential rupture (partial) 1/:		
S, J or K	4	11
L	0	1
Circumferential rupture (complete) 1/	0	1
Detached metal 2/	0	1
Projectile:		
Rotating band separation:		
Complete or partial separation having band seat serration imprint 3/	0	3
Separation having length greater than 0.5 inch without band seat serration imprint	1	6
Metal parts separation, except rotating band separation 4/	0	1

1/ Dwg. C7553663 (a detail of IEL 7259016) shall apply for classifying splits and ruptures in fired cartridge cases. If a longitudinal split or circumferential rupture (partial) extends into two or more defined areas, only the most severe defect criterion of Table II for the areas involved, shall apply. If a rupture results in separation of the cartridge case into two or more portions, the defect shall be classed as a complete circumferential rupture.

2/ Metal sheared or missing from the fired cartridge case exterior, such as a rim or neck shear, shall be classed as a defect. The lot shall not be penalized for metal shaved from the interior surface of the case neck in the crimped area.

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3/ For classification as a defect, there must be evidence either by recovery of the band, or portion thereof, or by hole(s) in the fragmentation screen(s). The lot shall not be penalized for normal band fringing, or the separation of a band fragment having no evidence of a band seat knurling imprint, and having length of 0.5 inch or less.

4/ Separation or breakup of projectile part(s), except rotating band separation, as witnessed by recovery of the part(s) or fragment(s), or hole(s) in the fragmentation screen(s) shall be classed as a defect.

4.3.4.3 Unlisted firing defects.-The lot shall be suspended and referred to the contracting officer for disposition, if a malfunction or casualty not covered by this specification occurring in any firing test, indicates that the product is unsuited for the purpose intended.

4.3.5 Packaging, packing and marking inspection.-Inspection for packaging, packing and marking shall be in accordance with MIL-STD-644, as applicable to the drawing.

4.3.6 Inspection equipment.-Index of Inspection Equipment List No. IEL 7259106 identifies the applicable inspection equipment list required to perform examination and tests prescribed herein. Equipment design shall be in accordance with code designation on the applicable IEL (reference is made to inspection equipment list for definition of coding format).

4.4 Test methods and procedures.

4.4.1 Projectile torque.-Equipment shall be as specified in IEL 7259106. Each cartridge of the test sample shall be marked with a light scratch extending axially across the rotating band onto the case neck. Torque specified shall be applied slowly to the projectile. Movement of the projectile with respect to the cartridge case, detectable by misalignment in the scratch mark shall be recorded.

4.4.2 Projectile extraction.-The method of test shall be as prescribed in AMCR 715-505, Volume 8.

4.4.3 Propellant contamination.-The propellant from each of the sample cartridges tested for projectile extraction shall be examined visually for contamination, as prescribed in AMCR 715-505, Volume 8.

4.4.4 Velocity, pressure and waterproofness.-The method of test shall be as prescribed in AMCR 715-505, Volume 8. Statistics necessary for testing of results against applicable acceptance criteria shall be computed. (see 6.6)

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4.4.5 Accuracy.-The method of test and measurement of targets shall be as prescribed in AMCR 715-505, Volume 8. The specified range may be shortened in decrements of 100 yards, to a minimum range of 200 yards. For ranges less than 600 yards the average mean radius shall be reduced in direct proportion to the reduction in range.

4.4.6 Action time.-The method of test shall be as prescribed in AMCR 715-505, Volume 8.

4.4.7 Function and casualty.-The method of test shall be as prescribed by AMCR 715-505, Volume 8. The test sample shall be fired in bursts of 50 cartridges. The weapon barrel shall be at ambient temperature at beginning of test and cooled to ambient after each 100 rounds. The cyclic rate of each burst shall not be less than 650 shots per minute (SPM) in the M3 gun. The cyclic rate for each burst shall be recorded.

4.4.8 Defective weapon.-If a test weapon fails to comply with the minimum cyclic rate during a normal burst operation or is found to be defective during or at the end of burst the test shall be suspended. The weapon shall be corrected or replaced and the test repeated in whole or in part, as indicated. Any firing defect that occurred shall be valid for lot penalty in accordance with Table II unless the defect is found to have been caused by the defective weapon.

5. PREPARATION FOR DELIVERY

5.1 Packing, Level A.-The type of ammunition pack shall be in accordance with instructions from the procuring activity. Linked ammunition shall be packed as specified in Dwg. F7258967, F7259188, F7259189 or F11075755. Bulk ammunition shall be packed as specified in Dwg. F7259123 or F7259108.

5.2 Marking.-Marking on the packed container shall comply with Dwg. D7258928 or D7259237, as applicable.

6. NOTES

6.1 Ordering data.-Invitation for bids and contracts or orders will specify the following:

6.1.1 Title, number, and date of this specification.

6.1.2 Place of inspection, if not at place of manufacture.

6.1.3 First article sample requirements. (see 4.2 and 6.2)

6.1.4 Detailed packing and marking instructions. (see 5.1 and 5.2)

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6.1.5 Provision for the supply, maintenance and disposition of mandatory ballistic test equipment for acceptance inspection purposes.

6.1.6 Provisions for the submission and approval of the manufacturing process. (see 3.2)

6.1.7 Place of firing tests for production lots and shipping instructions, if required, for production lot samples.

6.2 First article sample.-In order to minimize inspection costs, the firing tests of 4.2 will be performed after the sample has been provisionally accepted by the Government for all other requirements. Additional cartridges may be required by the test facility.

6.3 Process deviation.-A process deviation is defined as a change in the approved basic method of manufacture, or an operational change which may alter the metallurgical or physical properties of the item.

6.4 AQL's.-The optional use of AQL values for either individual defects or classes of defects, with individual major defect limitation, is intended to minimize inspection agency administrative burden which might result from an exclusive assignment of individual defect AQL's. The option also permits flexibility where sampling inspection for acceptance is integrated into the manufacturing process.

6.5 Contamination of explosive components.-Particular care should be exercised to avoid contamination of primers and propellant by oil, grease, or other foreign matter since such contamination can cause misfire(s) or increase action time and lot rejection.

6.6 Computations.

6.6.1 Standard deviation.-Where computation of a sample standard deviation is specified for determination of lot acceptance, the method of computation will be:

$$s = \sqrt{\frac{\Sigma(X_i - \bar{X})^2}{n}}$$ or equivalent

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6.6.2 Range plan.-Where a range plan is specified, the individual values will be recorded in subgroups of five in the order in which they are observed. The ranges for the subgroups will be added and the sum of the ranges will be divided by the number of subgroups to determine the average range (R).

Custodian:

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Preparing Activity:

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Project No. 1305-A604

4688

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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